

LAB TASK#6

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ROLL# : 22P-9326

SECTION: BCS-4C

QUESTION#1:

CODE:

```
#include <iostream>
#include <string>
using namespace std;

class Node
{
public:
string data;
Node *next;

Node(string val)
{
data = val;
next = NULL;
}
};

class Stack
{
public:
Node *top;
int limit;
int length;

Stack(int limit) : top(nullptr), limit(limit), length(0)
{
}
// constructor overloading
Stack() : top(nullptr), limit(0), length(0)
{
}

bool isEmpty()
{
return top == nullptr && length == 0;
}

bool isFull()
{
return length == limit;
}

void push(string val)
{
```

```
Node *n = new Node(val);
n->next = top;
top = n;
length++;
}

string pop()
{
if (isEmpty())
{
cout << "Stack is empty" << endl;
return "";
}
else
{
Node *temp = top;
string poppedVal = temp->data;
top = top->next;
delete temp;
length--;
return poppedVal;
}
}

~Stack()
{
while (!isEmpty())
{
pop();
}
}
};

class Browser
{

public: // all the data members are set public to avoid getter and setters
Stack forwardNavigationStack;
Stack backNavigationStack;
string currentURL;

Browser() : currentURL("") {}

~Browser()
{
}

void visit(const string &URL)
{
if (!currentURL.empty())
{
backNavigationStack.push(currentURL);
```

```
}

currentURL = URL;

if (forwardNavigationStack.isEmpty())
{
forwardNavigationStack.push(currentURL);
}

cout << "Visited " << currentURL << endl;
}

void back()
{
if (backNavigationStack.isEmpty())
{
cout << "Can't go back as there are no pages at back" << endl;
return;
}
else
{
currentURL = backNavigationStack.pop();
forwardNavigationStack.push(currentURL);
cout << "Back to: " << currentURL << endl;
}
}

void forward()
{
if (forwardNavigationStack.isEmpty())
{
cout << "Can't go forward as there are no pages at forward" << endl;
return;
}
else
{
currentURL = forwardNavigationStack.pop();
backNavigationStack.push(currentURL);
cout << "Forward to: " << currentURL << endl;
}
}

void display()
{
cout << "Current URL: " << currentURL << endl;
}
};

int main()
{
int choice;
string URL;
```

```
Browser B;

cout << "___WEB BROWSING___" << endl;
while (true)
{
cout << "1. Visit a page" << endl
<< "2. Navigate back" << endl
<< "3. Navigate forward" << endl
<< "4. Display current page" << endl
<< "5. Exit" << endl;
cin >> choice;

switch (choice)
{
case 1:
cout << "Enter URL: ";
cin >> URL;
B.visit(URL);
break;
case 2:
B.back();
break;
case 3:
B.forward();
break;
case 4:
B.display();
break;
case 5:
return 0;
default:
cout << "Invalid choice" << endl;
break;
}
}
}
```

```
Ishtiaq_web_browsing.cpp -o 22p9326_m
owsing
___WEB BROWSING___
1. Visit a page
2. Navigate back
3. Navigate forward
4. Display current page
5. Exit
1
Enter URL: yt.com
Visited yt.com
1. Visit a page
2. Navigate back
3. Navigate forward
4. Display current page
5. Exit
3
Forward to: yt.com
1. Visit a page
2. Navigate back
3. Navigate forward
4. Display current page
5. Exit
1
Enter URL: fb.com
Visited fb.com
1. Visit a page
2. Navigate back
```

```
2
Back to: yt.com
1. Visit a page
2. Navigate back
3. Navigate forward
4. Display current page
5. Exit
3
Forward to: yt.com
1. Visit a page
2. Navigate back
3. Navigate forward
4. Display current page
5. Exit
3
Forward to: fb.com
1. Visit a page
2. Navigate back
3. Navigate forward
4. Display current page
5. Exit
2
Back to: fb.com
1. Visit a page
2. Navigate back
3. Navigate forward
4. Display current page
5. Exit
```

```
1. Visit a page
2. Navigate back
3. Navigate forward
4. Display current page
5. Exit
3
Forward to: yt.com
1. Visit a page
2. Navigate back
3. Navigate forward
4. Display current page
5. Exit
3
Forward to: fb.com
1. Visit a page
2. Navigate back
3. Navigate forward
4. Display current page
5. Exit
3
Can't go forward as there are no page
1. Visit a page
2. Navigate back
3. Navigate forward
4. Display current page
5. Exit
5
mashaal@mashaal-Latitude-7480:~/Deskt
```

QUESTION#2

CODE:

```
#include <iostream>
#include <string>
using namespace std;

class Brackets
{
public:
bool isValid(const string &s)
{
int countRound = 0, countSquare = 0, countCurly = 0;

for (int i = 0; i < s.length(); ++i)
{
char ch = s[i];
if (ch == '(')
{
countRound++;
}
else if (ch == ')')
{
countRound--;
}
else if (ch == '[')
{
countSquare++;
}
else if (ch == ']')
{
countSquare--;
}
else if (ch == '{')
{
countCurly++;
}
else if (ch == '}')
{
countCurly--;
}

if (countRound < 0 || countSquare < 0 || countCurly < 0)
{
return false;
}
}
```

```
return countRound == 0 && countSquare == 0 && countCurly == 0;
}
};

int main()
{
Brackets B1;
string input;
int choice;

while (true)
{
cout << "1. Enter a string containing round, curly, and square brackets" << endl;
cout << "2. Exit" << endl;
cout << "Enter your choice: ";
cin >> choice;

if (choice == 1)
{
cout << "Enter the string: ";
cin.ignore();
getline(cin, input);

if (B1.isValid(input))
{
cout << "The string is valid." << endl;
}
else
{
cout << "The string is not valid." << endl;
}
}
else if (choice == 2)
{
return 0;
}
else
{
cout << "Invalid choice. Please try again." << endl;
}
}

return 0;
}
```

```
mashaal@mashaal-Latitude-7480:~/Des
Ishtiaq_brackets.cpp -o 22p9326_mas
1. Enter a string containing round,
2. Exit
Enter your choice: 1
Enter the string: ()
The string is valid.
1. Enter a string containing round,
2. Exit
Enter your choice: 1
Enter the string: (}
The string is not valid.
1. Enter a string containing round,
2. Exit
Enter your choice: 1
Enter the string: {()}
The string is valid.
1. Enter a string containing round,
2. Exit
Enter your choice: 2
```